Business Services Public Firm Analysis

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Executive summary

Business Services combines large software and digital platforms with many smaller and more specialized firms. The SIC 73 sample contains 30,610 distinct firm-year observations and 3,273 firms; only 64 appear in all 32 years from 1994 through 2025. This changing composition is central to interpreting industry averages.

The disclosure analysis matches 482 firms in 2020 and 510 in 2025. Data becomes the leading sector keyword under both word counts and aggregated TF-IDF. AI records the largest increase in both normalized keyword measures. These findings suggest a shift toward data and AI language, although filing artifacts and changes in the firms represented limit causal interpretation.

Microsoft has substantially greater 2020 revenue and ROA than its selected textually similar peers. Its later financial pattern nevertheless shows falling asset turnover: revenue nearly doubles between 2020 and 2025, while assets more than double. We recommend making new cloud and AI capacity investment conditional on demonstrated utilization and incremental returns.

Industry selection and sample construction

We select Business Services using major_group 73 in major_groups.csv. The code converts SIC to a numeric value and retains observations for which floor(SIC / 100) equals 73. This applies the industry classification recorded for each firm-year rather than assuming that a firm remains in one sector throughout its history.

Sample measure	Result
Distinct gvkey and fyear pairs	30,610
Distinct firms identified by gvkey	3,273
Firms observed in every year from 1994 to 2025	64

Firm-year counts remove duplicate gvkey and fyear pairs. Complete coverage requires the set of observed years to contain every year from 1994 to 2025. The 64 fully observed firms represent approximately 1.96% of the firms in the sector sample. The panel is therefore highly unbalanced; entries, exits and industry reclassification can change the averages.

Preliminary quantitative analysis

Highest stock prices in 2020 and 2025

For each year, we exclude missing prcc_c values and rank the remaining observations in descending order. Prices retain the source units. A high price per share does not establish a high market capitalization, and comparisons can be affected by stock splits and currency conventions.

2020 firm	Price	2025 firm	Price
Alphabet	1,752.64	Constellation Software	2,399.743
Constellation Software	1,303.04	Fair Isaac	1,690.62
Shopify	1,131.95	United Rentals	809.32
CoStar Group	924.28	AppLovin	673.82
Trade Desk	801.00	Intuit	662.42
ServiceNow	550.43	Meta Platforms	660.09
Fair Isaac	511.04	Spotify Technology	580.71
Adobe	500.12	Mastercard	570.88
Paycom Software	452.25	CACI International	532.81
Tyler Technologies	436.52	S&P Global	522.59

Only Constellation Software and Fair Isaac appear in both top-ten lists. The turnover describes nominal share-price rankings, rather than a ranking of total shareholder returns or firm value.

Sales geography and industry trends

Highest historical sales by distinct firm

We sort all nonmissing sales observations from highest to lowest, retain the first observation for each gvkey, and then select ten firms. Each row below is a different firm at its historical peak sales; this avoids filling the ranking with multiple years of the same company.

Firm	Peak year	Sales in source units
Alphabet	2025	402,836.000
Microsoft	2025	281,724.000
Meta Platforms	2025	200,966.000
IBM	2011	106,916.000
Oracle	2025	67,357.000
Fujitsu	2010	54,717.315
NEC	1999	48,460.650
SAP	2025	43,207.817
Salesforce	2025	41,525.000
Lucent Technologies	1999	38,303.000

Top ten locations

Location	Unique firms	Location	Unique firms
USA	2,684	HKG	39
CHN	152	SGP	33
CAN	80	JPN	22
ISR	76	IRL	15
GBR	56	NLD	11

The code counts distinct gvkey values within each location over the full sample. USA accounts for 2,684 firms, approximately 82.0% of the 3,273 distinct firms. A firm can contribute to more than one location if its recorded location changes. US listing also does not imply US domicile.

Largest stock price decline during the financial crisis

For firms with observations in both 2007 and 2008 and a positive 2007 price, the drop is 100 × (price in 2007 − price in 2008) / price in 2007. ACCESS NEWSWIRE INC (gvkey 183139) has the largest observed drop: from 1.25 to 0.0071, or 99.432%. This is an observed price decline over the crisis interval; the calculation alone cannot attribute the entire decline to the financial crisis.

Average stock price and return on assets

Annual stock price is the arithmetic mean of available prcc_c observations in each year. USA ROA is calculated for each firm-year as net_income / asset, after replacing zero assets with missing values, then averaged across available USA observations and multiplied by 100.

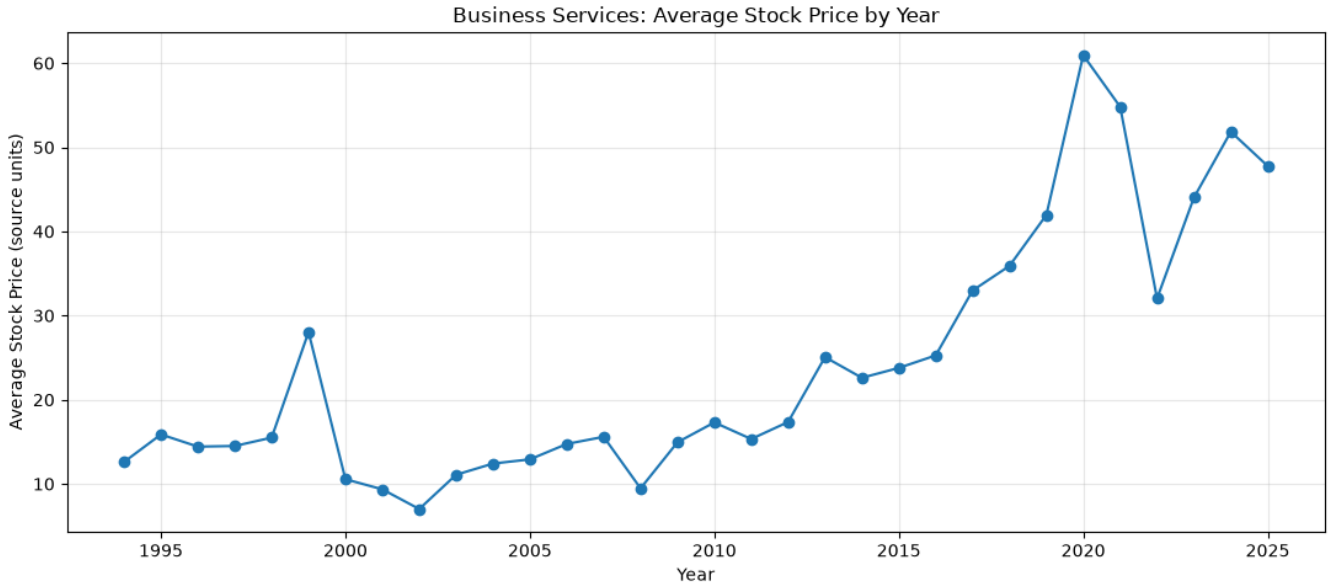


Figure 1 · Annual mean stock price for Business Services

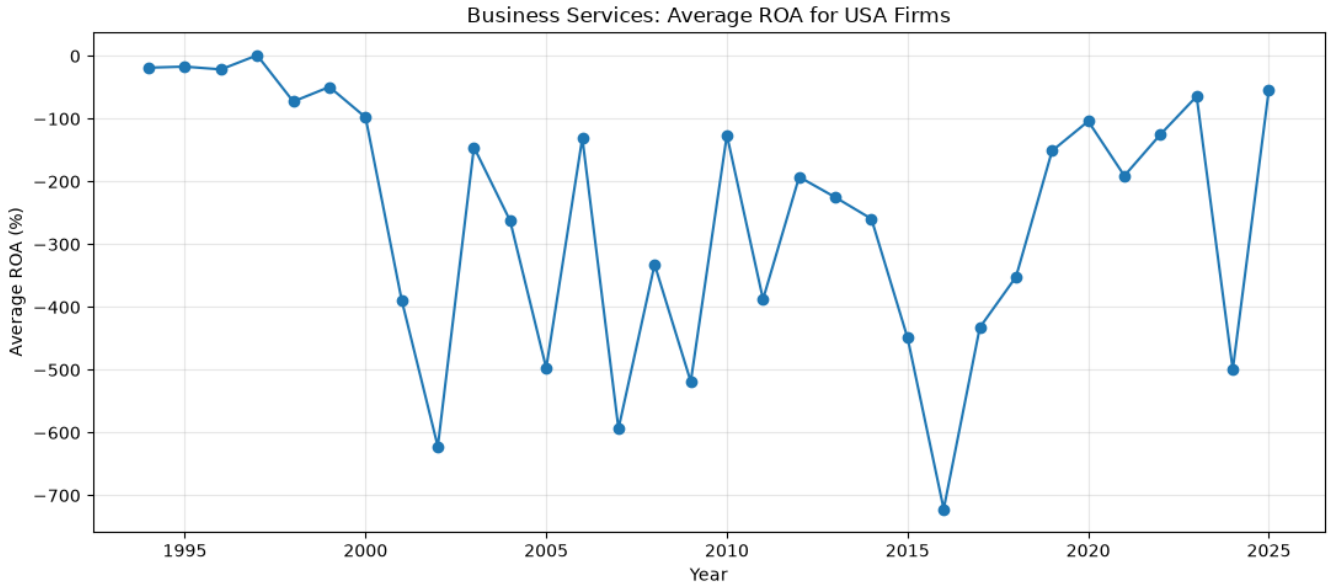


Figure 2 · Annual mean ROA for USA Business Services firms

The mean price chart rises markedly in later years but has substantial fluctuations. The ROA series is negative in almost all displayed years and includes extreme negative years. An equal-weighted mean of ratios can be driven by small denominators and loss-making firms. These averages should not be read as a representative firm's return or an investable sector return. A median and an asset-weighted ROA would be useful robustness checks.

Text preparation and keyword analysis

For each filing year, we lowercase Item 1 text, replace Unicode punctuation with spaces, retain alphabetic tokens of at least two characters, and remove NLTK English stop words. Missing text becomes an empty string. The final pipeline also excludes numbers and single-letter fragments; it does not stem or lemmatize, so customer and customers remain different terms.

Merge key and one document per firm

We use gvkey, a common firm identifier, instead of company name, which can change or have inconsistent spelling. For each year we first select that year's SIC 73 financial records and require one row per gvkey. Exact duplicate filings are removed, and distinct cleaned filing texts are combined into one document per firm. An inner join on gvkey with one-to-one validation retains firms present in both sources. This prevents a filing from being multiplied across 32 financial years.

Measure	2020	2025
Full sample usable filing documents	5,152	4,536
Matched sector firms	482	510
Sector firms with usable text	482	510
Firms present in both sector text samples	306	306

CountVectorizer constructs a document-term matrix. The code produces ten keywords per firm by raw frequency and by TF-IDF, sorting score ties alphabetically. TF-IDF uses smoothed inverse document frequency and L2 normalization: idf = log((1 + N)/(1 + document frequency)) + 1. Sector wordclouds use column sums of counts or normalized TF-IDF scores. Counts emphasize repeated words and longer filings; TF-IDF gives greater importance to terms that distinguish individual filings.

Microsoft keyword example for 2020

Rank	Count keyword	Count	TF-IDF keyword	Score
1	services	106	microsoft	.3069
2	microsoft	68	windows	.2524
3	products	65	services	.2187
4	cloud	59	cloud	.1740
5	business	54	xbox	.1636
6	software	46	gaming	.1634
7	customers	45	server	.1612
8	devices	44	devices	.1430
9	solutions	44	products	.1430
10	including	43	president	.1314

Counts emphasize broad services; TF-IDF highlights product names. President indicates residual administrative text.

Sector keyword shifts

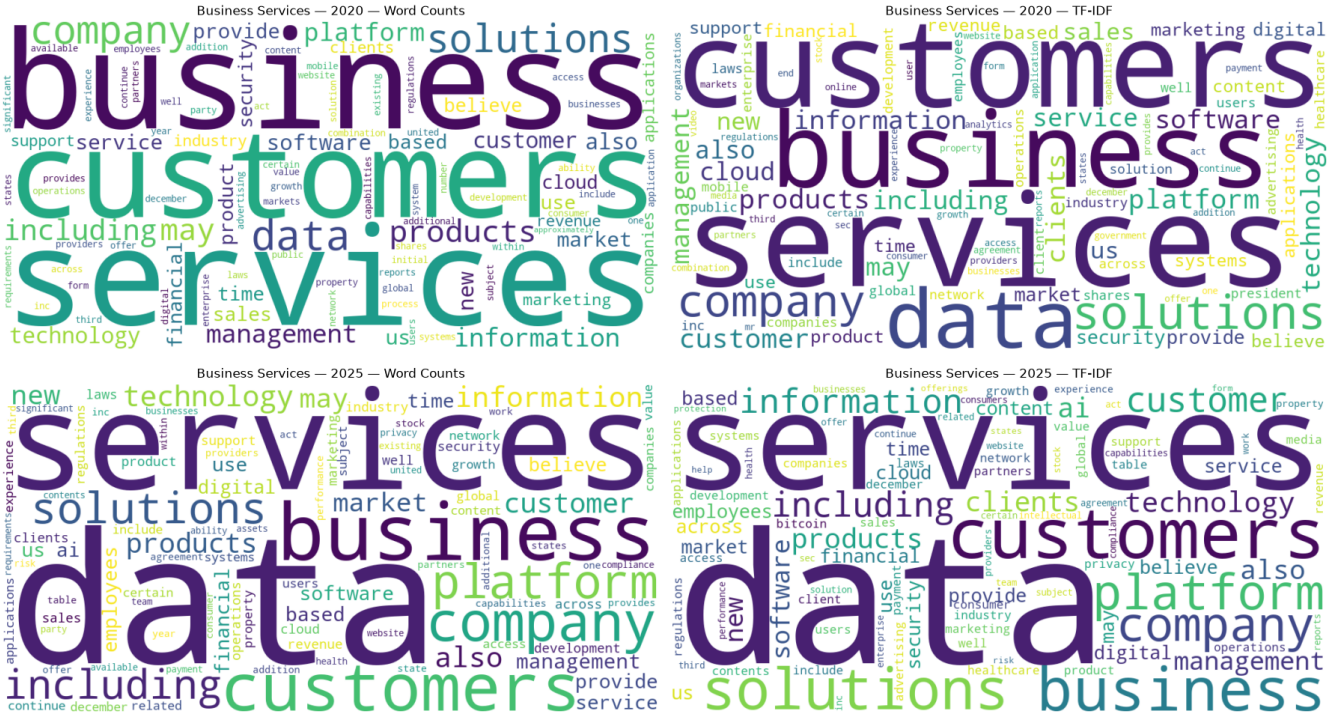


Figure 3 · Word counts and aggregated TF-IDF for 2020 and 2025

Both methods identify data as the leading 2025 term. In 2020, services ranks first in both methods, while data ranks fourth. Platform moves into the top ten counts and rises from eighth to sixth in aggregated TF-IDF. Services, customers, business and solutions remain prominent, indicating continuity in the sector's customer and service orientation.

Keyword	Count share 2020	Count share 2025	Change in pp
ai	0.033%	0.282%	+0.24858
data	0.602%	0.714%	+0.11134
platform	0.395%	0.492%	+0.09689
business	0.779%	0.582%	−0.19697
services	0.782%	0.639%	−0.14350

AI also has the largest TF-IDF share increase, +0.16054 percentage points. This is consistent with greater emphasis on AI in disclosure language, but does not measure AI revenue or adoption. Bitcoin and mining also gain TF-IDF share, potentially reflecting particular firms or changes in sector composition. Rising table and contents terms indicate extraction artifacts rather than economic trends.

Only 306 firms occur in both samples, so aggregate changes mix within-firm language changes with entry and exit. Year-specific TF-IDF models also use different document frequencies. Share normalization addresses corpus size, but does not remove these comparability issues. A matched-firm analysis would strengthen the interpretation.

Word embeddings and semantic comparison

A custom NumPy implementation trains skip-gram Word2Vec with negative sampling on the full cleaned filing sample for each year, rather than only SIC 73 firms. Settings are 50 dimensions, context window up to five tokens, minimum frequency five, five negative samples, three epochs, batch size 512, initial learning rate 0.025 and seed 42. Frequent words are subsampled at threshold 0.0001 and the negative distribution is proportional to frequency raised to 0.75.

The resulting vocabularies contain 55,630 words in 2020 and 54,646 in 2025. Mean training loss declines from 2.1871 to 2.0725 and from 2.1780 to 2.0712 respectively. Declining loss indicates optimization progress; it does not independently validate semantic quality. Software, services and customers are selected as representative terms visible in the sector wordclouds.

Year and query	Five nearest words with cosine similarity
2020 software	hardware .8460; middleware .8123; dynatrace .8041; storsight .7943; certicom .7928
2025 software	hardware .8857; spotlightaitm .7382; firmware .7319; efpga .7315; rtls .7300
2020 services	rbwm .7980; optumhealth .7802; qhr .7603; iwco .7599; presort .7574
2025 services	zendit .7013; nemt .6991; firstech .6979; consortiums .6923; service .6873
2020 customers	ccas .7602; customer .7432; shopfast .7287; finapps .7285; msps .7141
2025 customers	billers .7520; users .7426; businesses .7337; lifecycles .7288; demanded .7248

Hardware remains the closest neighbor to software in both years; it is the only shared top-five software neighbor. Services and customers have no top-five overlap between years. The 2025 customers neighborhood contains more general user and business terms, while the software neighborhood includes firmware and technical acronyms. These are descriptive shifts in contextual association, not proof that firms changed their products.

Several neighbors are company names, abbreviations or unusual tokens. Because training uses all industries, services can acquire associations from healthcare or financial services rather than the selected sector. Separate yearly models are not aligned; nearest-neighbor overlap can be compared, but raw vector coordinates cannot be subtracted to measure semantic change. Limited epochs, stochastic training and text quality may also affect the neighbors.

Data quality implications

Some matched 2020 filing examples are inconsistent with their displayed company labels: the AUTOMATIC DATA PROCESSING example describes stevioside, while WORLDS INC is dominated by cannabis language. These examples warrant checking gvkey against original filing identifiers and dates before treating all firm-level text matches as reliable. One-to-one validation prevents duplicate rows but cannot verify that a source identifier is correct.

Microsoft and its selected peers

We focus on Microsoft's position relative to firms with similar 2020 business descriptions. The final peer-selection code uses cosine similarity between full TF-IDF document vectors. It does not sum Word2Vec keyword embeddings. This is a valid alternative for the open-ended firm analysis and measures shared disclosure vocabulary, rather than direct product-market rivalry.

Selected peer	TF-IDF cosine similarity
F5	.4035
Salesforce	.3980
Open Text	.3868
VMware	.3493
Progress Software	.3476

Cloud, software and enterprise themes help explain the peer set. Microsoft's cloud TF-IDF score is .1740 versus a peer mean of .1905; platform is .0384 versus .0750. These differences describe emphasis within documents, not capability, market share or product quality. The peers span complementary as well as competing activities.

Comparable financial position in 2020

Firm	Sales	Assets	ROA	Sector share
Microsoft	143,015.000	301,311.000	14.696%	9.504%
Salesforce	21,252.000	66,301.000	6.142%	1.412%
VMware	11,767.000	29,016.000	7.093%	0.782%
Open Text	3,109.736	10,234.822	2.289%	0.207%
F5	2,350.822	4,677.920	6.572%	0.156%
Progress Software	442.150	1,041.782	7.652%	0.029%

ROA is 100 × net income / positive ending assets for every firm. Sector share is a sample sales proxy: firm sales divided by total nonnegative available sales for SIC 73 firms in 2020. It is not global software or cloud market share. Financial amounts retain the CSV's units; currency and scale comparability should be confirmed before interpreting absolute differences.

Microsoft's revenue is about 46 times the selected competitor median of 3,109.736. Its ROA is 8.124 percentage points above the peer median of 6.572%. Microsoft therefore combines scale with stronger measured asset profitability in this selected comparison. The result does not establish that scale causes higher ROA or that these five firms exhaust Microsoft's competitive environment.

Strategic recommendation for Microsoft

Make incremental cloud and AI capacity investment conditional on demonstrated utilization and expected incremental returns. Microsoft's strong peer position provides room to invest, while the decline in asset turnover makes disciplined deployment particularly relevant.

Year	Sales	Assets	Net margin	Turnover	ROA
2020	143,015	301,311	30.962%	.475	14.696%
2021	168,088	333,779	36.452%	.504	18.357%
2022	198,270	364,840	36.686%	.543	19.937%
2023	211,915	411,976	34.146%	.514	17.564%
2024	245,122	512,163	35.956%	.479	17.209%
2025	281,724	619,003	36.146%	.455	16.451%

From 2020 to 2025, sales grow about 97.0% and assets about 105.4%. From the 2022 peak, ROA falls by 3.486 percentage points while asset turnover declines from .543 to .455. Net profit margin changes only modestly, from 36.686% to 36.146%. The accounting identity ROA = net profit margin × asset turnover therefore points primarily to lower asset utilization relative to revenue in this interval. Total assets include more than cloud infrastructure, so the data do not isolate the cause of the decline.

Implement the recommendation through staged capacity approvals supported by contracted demand, utilization forecasts and a consistent incremental return hurdle. Review utilization, incremental revenue per unit of invested capital and ROA quarterly, and delay further expansion where demand persistently trails forecasts. This is one capital-allocation recommendation; the analysis does not support a numerical hurdle or claim that all AI investment should be reduced.

Supporting historical context

Microsoft's largest observed annual stock-price increase is 56.801% in 2023 and its largest decline is 62.848% in 2000. Its largest revenue increase is 46.050% in 1996 and largest decline is 8.827% in 2016. These identify periods for investigation, but the dataset does not establish event-specific causes.

Stock-price levels correlate strongly with revenue (.854) and assets (.862), but annual stock-price changes correlate weakly and negatively with revenue growth (−.169) and asset growth (−.156). Revenue and asset levels correlate at .986, versus .624 for their growth rates. Common long-term trends can inflate level correlations. These associations neither demonstrate causality nor support forecasting stock returns from revenue alone.

Conclusion

The sector's disclosures increasingly emphasize data and AI, while the financial sample remains broad and compositionally unstable. Microsoft's scale and peer ROA advantage are clear in the 2020 comparison, but its recent asset turnover warrants attention. Investment tied to demand and returns is the most directly supported strategic response.

References and analytical notes

Project materials

The Group 26 project notebook contains the executed Python analysis. The notebook supplied the quantitative results, keyword statistics, Word2Vec training outputs and figures. The final asset-based comparison supersedes the earlier exploratory cells that did not recognize the asset column.

Course datasets referenced by the notebook: major_groups.csv; public_firms.csv; 2020_10K_item1_full.csv; 2025_10K_item1_full.csv. Financial quantities are reported in their supplied units; no external financial figures are introduced. Also, course materials from Gene Lee's class are used.

Generative AI disclosure

The team used ChatGPT (identified in the course submission as Sol Light, OpenAI) to develop the initial Python code, assist with coding for tasks A–G, and draft the report.

Interpretation and reproducibility

The quantitative pipeline uses pandas for filtering, distinct counts, annual aggregation and ranking, with matplotlib for figures. The text pipeline uses NLTK stop words, scikit-learn count and TF-IDF matrices, WordCloud for visual summaries, and a NumPy Word2Vec implementation. No code screenshots are needed to understand the procedures or results.

Missing prices and sales are excluded from the corresponding rankings. USA sector-average ROA removes zero assets; the final Microsoft comparison uses positive assets. Small asset denominators and extreme observations can dominate arithmetic mean ROA. A fixed cohort, median ROA and asset-weighted ROA would help evaluate robustness.

The sector aggregates combine firms with diverse business models and different reporting calendars. Historical company labels may differ from names used in a given year. Nominal stock prices are not total returns, and financial series are not adjusted here for inflation. Year labels and financial units follow the supplied course data.

Keyword outputs are computed for every matched firm; the report presents a complete Microsoft example and sector summaries to keep the discussion readable. Exact duplicate filings are removed, but concatenation of distinct filings can still overweight firms with more disclosure text. Firm names and boilerplate remain in the vocabulary.

The recommended investment discipline is an analytical proposal based on consolidated historical results. Establishing product-market shares, segment returns or the causes of financial changes would require additional evidence beyond this analysis.